Exploratory Data Analysis on a Cars Dataset

Cars EDA Minor Project

Student	Dhairya Chovatiya
Project Type	Individual EDA Project
Dataset	Vehicle Dataset from CarDekho
Dataset Source	Kaggle — Nehal Birla
Tools	Python, Pandas, NumPy, Matplotlib, Seaborn
Submission	Code file + PDF report

Project objective: clean, explore, visualise and explain meaningful patterns in used-car data. The assignment does not require a machine-learning model.

1. Objective and Dataset Overview

The project performs a complete EDA of used-car listings. It examines data structure and quality, descriptive statistics, categorical distributions, numerical relationships, outliers and evidence-based insights.

Main questions

• Which fuel types and brands are most common?

• How are selling prices distributed?

• Does vehicle year relate to selling price?

• Does kilometres driven relate to selling price?

• How common are manual and automatic cars?

• How does ownership history relate to selling price?

Dataset

The Kaggle Vehicle Dataset from CarDekho contains 4,340 rows and 8 columns: name, year, selling_price, km_driven, fuel, seller_type, transmission and owner. It contains three numerical variables and five categorical variables.

Feature	Type	Description
name	Categorical	Car/model name
year	Numerical	Vehicle year
selling_price	Numerical	Selling price in INR
km_driven	Numerical	Distance driven
fuel	Categorical	Fuel type
seller_type	Categorical	Seller category
transmission	Categorical	Manual / Automatic
owner	Categorical	Ownership history

Source: Kaggle Vehicle Dataset from CarDekho.

2. Data Quality and Cleaning

Check	Result / Decision
Initial size	4,340 rows × 8 columns
Missing values	0 in all eight columns
Data types	3 numerical + 5 categorical
Duplicate check	Performed using pandas and reported by code
Text cleaning	Trim whitespace from text fields
Numeric cleaning	Convert year, selling_price and km_driven safely
Outlier method	IQR rule used for selling_price and km_driven
Outlier decision	Extreme values retained unless clearly invalid
Rows after basic cleaning	4,340; no missing-value rows required removal

Cleaning rationale

The assignment requires missing values and outliers to be treated only when justified. Luxury cars and very high-use vehicles can be legitimate records, so the project identifies outliers rather than automatically deleting them. Exact duplicate count is printed by the submitted code so the student can explain the result from the actual file used.

Before/after cleaning

The core dataset remains 4,340 rows after basic cleaning because no missing values were present and the cleaning operations mainly correct formatting and data types. Duplicate and outlier decisions are kept transparent in the code.

The submitted code also creates a derived car_age variable and performs all required descriptive and visual analysis.

3. Descriptive Analysis

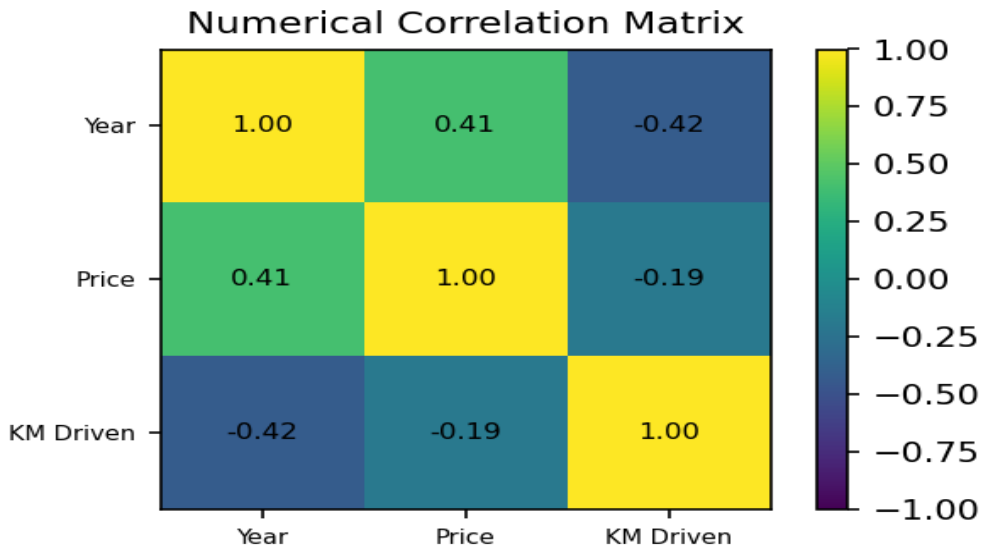
Variable	Count	Mean	Median	Min	Max	Std. Dev.
Year	4,340	2013.09	2014	1992	2020	4.22
Selling Price (INR)	4,340	504,127	350,000	20,000	8,900,000	578,549
KM Driven	4,340	66,216	60,000	1	806,599	46,644

Interpretation

Selling price has a mean of about ■5.04 lakh but a median of ■3.50 lakh, showing a strongly right-skewed distribution. Kilometres driven also has a large spread, from 1 km to 8,06,599 km.

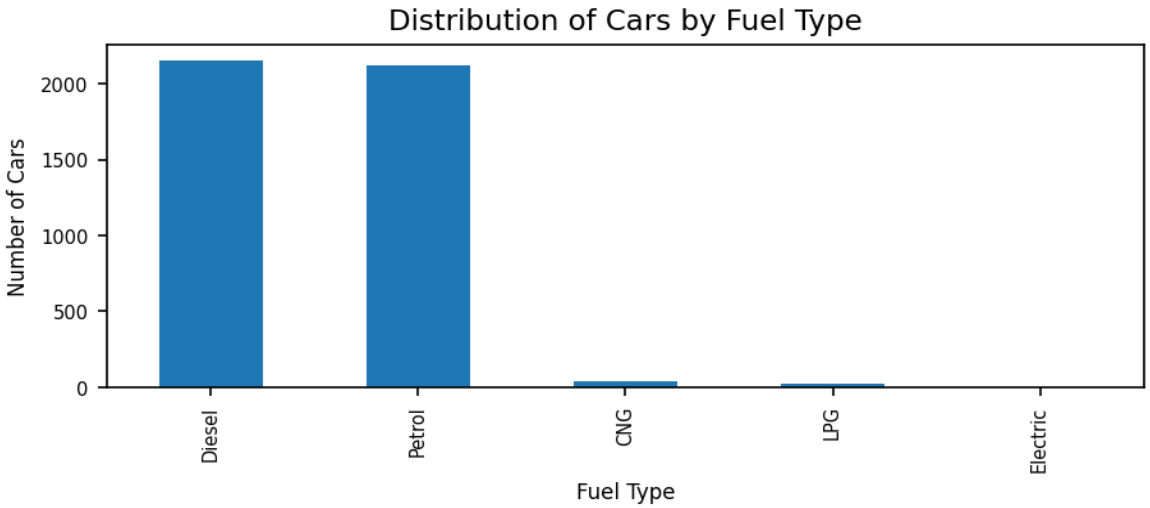
Correlation results

Year vs selling price ≈ +0.41; km driven vs selling price ≈ –0.19; year vs km driven ≈ –0.42. These values indicate that newer vehicles generally have higher prices, while kilometres driven alone has a comparatively weak relationship with price.

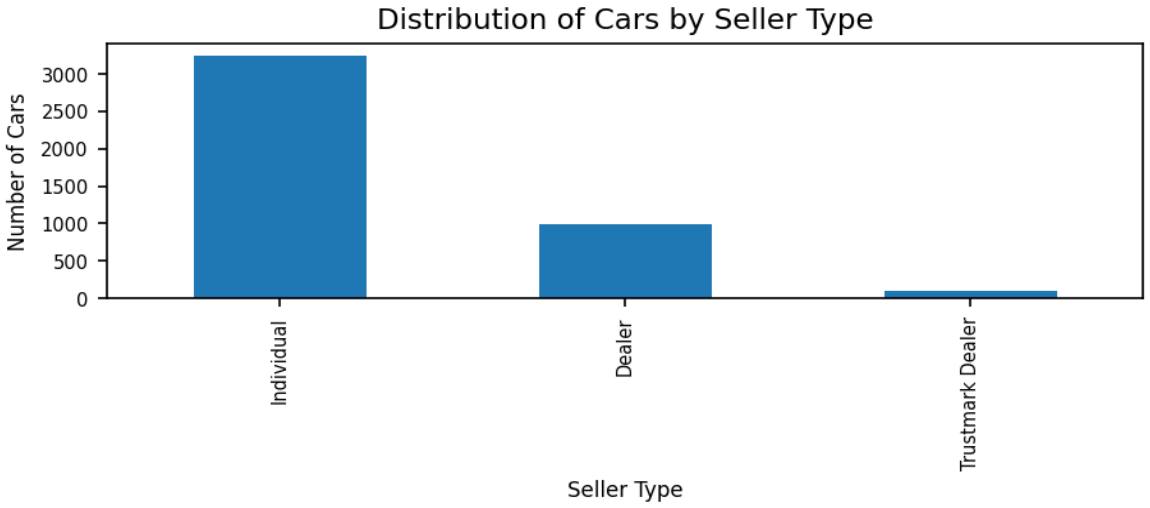


Observation: the correlation pattern supports both univariate and multivariate EDA. Correlation indicates association, not causation.

4. Visual EDA

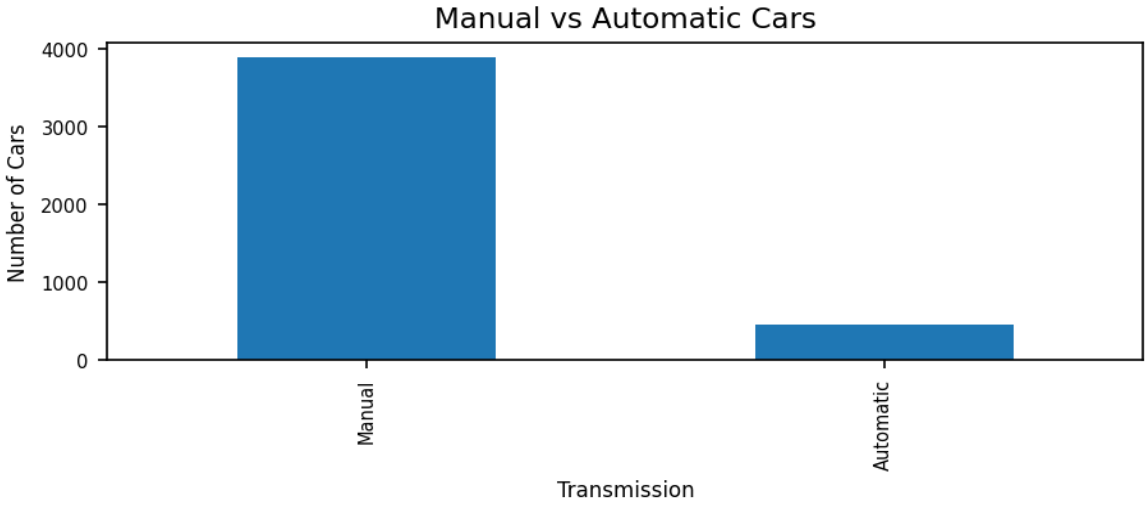


Observation: Diesel and Petrol dominate the dataset; CNG and LPG are much smaller groups and Electric appears only once.

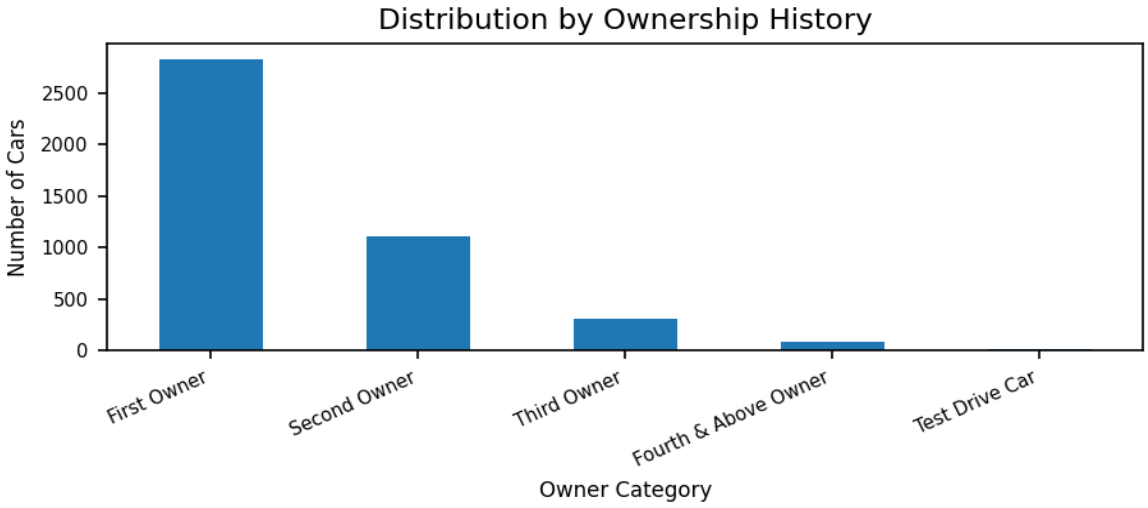


Observation: Individual sellers are the largest group by a wide margin, followed by Dealers; Trustmark Dealer listings are comparatively rare.

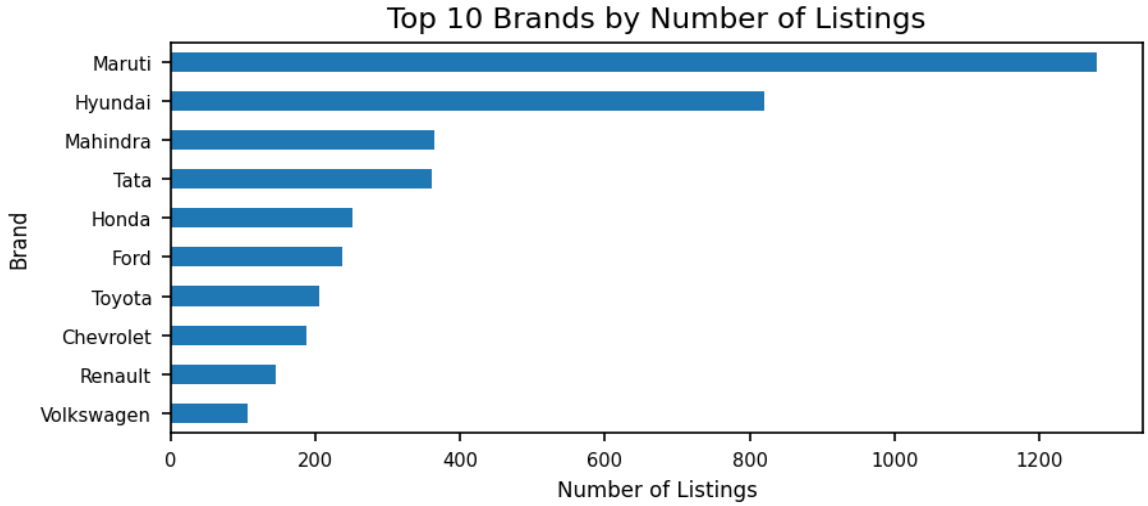
4. Visual EDA — Continued



Observation: Manual cars strongly dominate the dataset, so automatic cars are a relatively small segment.

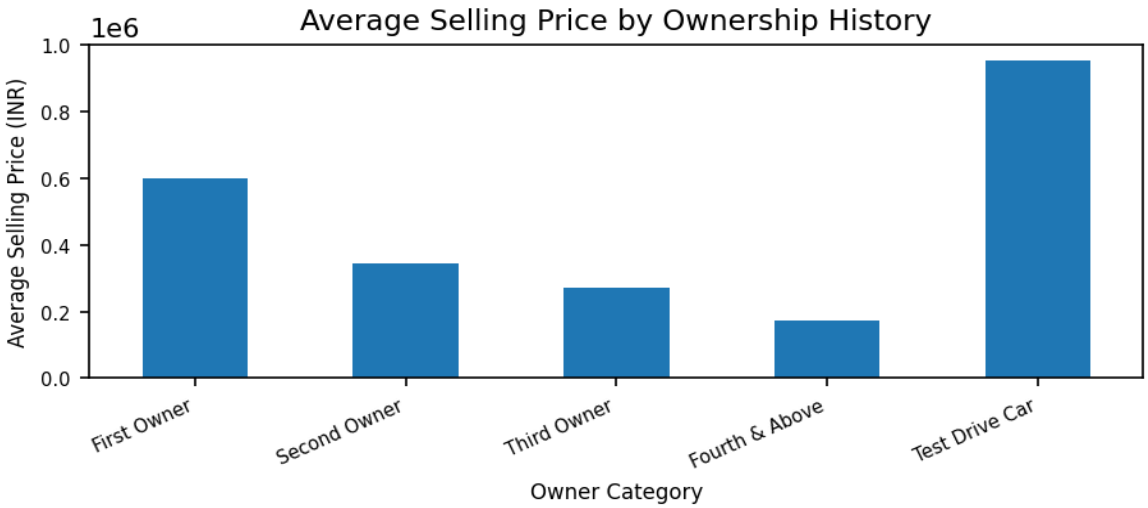


Observation: First-owner vehicles are the largest ownership group, followed by second-owner vehicles.



Observation: Maruti is the most frequently listed brand, followed by Hyundai; the two are substantially ahead of the other brands.

5. Multivariate / Group Analysis



Observation: Average selling price differs across ownership categories. Test-drive cars have the highest average in the available summary, while fourth-and-above-owner cars have the lowest.

Key Findings

1. Petrol (2,123) and Diesel (2,153) are overwhelmingly the dominant fuel categories.

2. Manual transmission accounts for 3,892 of the 4,340 listings, so the dataset is strongly skewed toward manual vehicles.

3. Individual sellers account for 3,244 listings, making them the largest seller category.

4. Maruti is the most frequently listed brand (1,280), followed by Hyundai (821).

5. Year and selling price have a moderate positive correlation of about +0.41, indicating that newer cars generally have higher selling prices.

6. Kilometres driven and selling price have a weak negative correlation of about –0.19, so mileage alone does not strongly explain selling price.

7. The large difference between mean selling price (■5.04 lakh) and median selling price (■3.50 lakh) indicates a right-skewed distribution with expensive vehicles pulling the mean upward.

6. Limitations and Conclusion

Limitations

• The dataset represents used-car listings and should not be treated as a complete census of the Indian used-car market.

• The dataset does not include every feature that can affect price, such as city, condition, service history, insurance status, accident history, or detailed engine specifications.

• Observational relationships are associations, not proof of causation.

• Very rare categories, such as Electric and Test Drive Car, should be interpreted cautiously because their sample sizes are small.

• Duplicate listings and extreme values are explicitly checked by the code; they should not be removed automatically without understanding their source.

Conclusion

This EDA shows a used-car market dominated by Petrol/Diesel vehicles, manual transmission, individual sellers, first-owner cars and Maruti/Hyundai listings. Newer model years are associated with higher selling prices, while kilometres driven has a weaker negative relationship with price. The project demonstrates a complete EDA workflow without using a machine-learning model.

Future scope

A larger CarDekho version containing mileage, engine, maximum power, torque and seats could support deeper analysis. Location, vehicle condition and service history would also improve future studies. Predictive modelling could be added later, but it is outside this assignment.

References

1. Kaggle — Vehicle Dataset from CarDekho, Nehal Birla and collaborators.

2. Public analyses of the same 4,340-row dataset were used only to cross-check descriptive values and category counts; the submitted Python file performs the EDA directly on the dataset.

Submission package: exactly one Python code file and one PDF report.